```
mysql>  CREATE DATABASE CollegeJoinDB;

Query OK, 1 row affected (0.09 sec)


mysql>  USE CollegeJoinDB;

Database changed

mysql>  CREATE TABLE Departments ( DeptID INT PRIMARY KEY,DeptName VARCHAR(50),
Location VARCHAR(50));

Query OK, 0 rows affected (0.10 sec)


mysql>  CREATE TABLE Employees (EmpID INT PRIMARY KEY,EmpName VARCHAR(50),DeptID
INT,Salary DECIMAL(10,2),City VARCHAR(50));

Query OK, 0 rows affected (0.02 sec)


mysql> CREATE TABLE SalaryGrades (Grade CHAR(2) PRIMARY KEY,MinSalary
DECIMAL(10,2),MaxSalary DECIMAL(10,2));

Query OK, 0 rows affected (0.05 sec)


mysql>  INSERT INTO Departments VALUES (10,'Computer Science','Block
A'),(20,'Pharmacy','Block B'),(30,'Management','Block C'),(40,'Civil Engineering','Block D');

Query OK, 4 rows affected (0.04 sec)

Records: 4  Duplicates: 0  Warnings: 0


mysql> > INSERT INTO Employees VALUES
(101,'Aarav',10,65000,'Delhi'),(102,'Priya',20,72000,'Mumbai'),(103,'Rahul',10,48000,'Delhi')
,(104,'Sneha',30,91000,'Pune'),(105,'Vikram',20,55000,'Jaipur'),(106,'Neha',50,80000,'Bengal
uru'),(107,'Kabir',30,43000,'Pune');

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near '> INSERT INTO
Employees VALUES (101,'Aarav',10,65000,'Delhi'),(102,'Priya',20,72' at line 1
```

```
mysql> INSERT INTO Employees VALUES
(101,'Aarav',10,65000,'Delhi'),(102,'Priya',20,72000,'Mumbai'),(103,'Rahul',10,48000,'Delhi')
,(104,'Sneha',30,91000,'Pune'),(105,'Vikram',20,55000,'Jaipur'),(106,'Neha',50,80000,'Bengal
uru'),(107,'Kabir',30,43000,'Pune');

Query OK, 7 rows affected (0.04 sec)

Records: 7  Duplicates: 0  Warnings: 0


mysql> INSERT INTO SalaryGrades VALUES
('G1',0,49999),('G2',50000,69999),('G3',70000,89999),('G4',90000,120000);

Query OK, 4 rows affected (0.03 sec)

Records: 4  Duplicates: 0  Warnings: 0


mysql>  SHOW TABLES;

+-------------------------+

| Tables_in_collegejoindb |

+-------------------------+

| departments             |

| employees               |

| salarygrades            |

+-------------------------+

3 rows in set (0.02 sec)


mysql>  SELECT * FROM Departments;

+--------+------------------+----------+

| DeptID | DeptName         | Location |

+--------+------------------+----------+

|     10 | Computer Science  | Block A  |

|     20 | Pharmacy         | Block B  |

|     30 | Management       | Block C  |
```

```
|    40 | Civil Engineering | Block D  |
+--------+-------------------+----------+
4 rows in set (0.00 sec)

mysql>  SELECT * FROM Employees;
+-------+---------+--------+----------+-----------+
| EmpID | EmpName | DeptID | Salary   | City      |
+-------+---------+--------+----------+-----------+
|   101 | Aarav   |     10 | 65000.00 | Delhi     |
|   102 | Priya   |     20 | 72000.00 | Mumbai    |
|   103 | Rahul   |     10 | 48000.00 | Delhi     |
|   104 | Sneha   |     30 | 91000.00 | Pune      |
|   105 | Vikram  |     20 | 55000.00 | Jaipur    |
|   106 | Neha    |     50 | 80000.00 | Bengaluru |
|   107 | Kabir   |     30 | 43000.00 | Pune      |
+-------+---------+--------+----------+-----------+
7 rows in set (0.00 sec)

mysql>  SELECT * FROM SalaryGrades;
+-------+-----------+-----------+
| Grade | MinSalary | MaxSalary |
+-------+-----------+-----------+
| G1    |      0.00 |  49999.00 |
| G2    |  50000.00 |  69999.00 |
| G3    |  70000.00 |  89999.00 |
| G4    |  90000.00 | 120000.00 |
+-------+-----------+-----------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT e.EmpName, d.DeptName, d.Location
    -> FROM Employees e
    -> INNER JOIN Departments d
    -> ON e.DeptID = d.DeptID;
+---------+------------------+----------+
| EmpName | DeptName         | Location |
+---------+------------------+----------+
| Aarav   | Computer Science | Block A  |
| Priya   | Pharmacy         | Block B  |
| Rahul   | Computer Science | Block A  |
| Sneha   | Management       | Block C  |
| Vikram  | Pharmacy         | Block B  |
| Kabir   | Management       | Block C  |
+---------+------------------+----------+
6 rows in set (0.02 sec)
```

```
mysql> SELECT e.EmpID, e.EmpName, e.DeptID, d.DeptName FROM Employees e INNER
JOIN Departments d ON e.DeptID = d.DeptID;
+-------+---------+--------+------------------+
| EmpID | EmpName | DeptID | DeptName         |
+-------+---------+--------+------------------+
|   101 | Aarav   |     10 | Computer Science |
|   102 | Priya   |     20 | Pharmacy         |
|   103 | Rahul   |     10 | Computer Science |
|   104 | Sneha   |     30 | Management       |
|   105 | Vikram  |     20 | Pharmacy         |
|   107 | Kabir   |     30 | Management       |
```

```
+-------+---------+--------+------------------+
```

6 rows in set (0.00 sec)

```
mysql> SELECT e.EmpID, e.EmpName, d.DeptName FROM Employees e INNER JOIN
Departments d ON e.DeptID = d.DeptID WHERE d.DeptName = 'Pharmacy';
+-------+---------+----------+
| EmpID | EmpName | DeptName |
+-------+---------+----------+
|   102 | Priya   | Pharmacy |
|   105 | Vikram  | Pharmacy |
+-------+---------+----------+
```

2 rows in set (0.03 sec)

```
mysql> SELECT e.EmpName, d.DeptName, e.Salary FROM Employees e INNER JOIN
Departments d ON e.DeptID = d.DeptID WHERE e.Salary > 60000;
+---------+------------------+----------+
| EmpName | DeptName         | Salary   |
+---------+------------------+----------+
| Aarav   | Computer Science | 65000.00 |
| Priya   | Pharmacy         | 72000.00 |
| Sneha   | Management       | 91000.00 |
+---------+------------------+----------+
```

3 rows in set (0.00 sec)

```
mysql> SELECT e.EmpName, d.DeptName FROM Employees e INNER JOIN Departments d
ON e.DeptID = d.DeptID;
+---------+------------------+
| EmpName | DeptName         |
+---------+------------------+
```

| Aarav | Computer Science |

| Priya | Pharmacy |

| Rahul | Computer Science |

| Sneha | Management |

| Vikram | Pharmacy |

| Kabir | Management |

+---------+------------------+

6 rows in set (0.00 sec)

mysql> SELECT e.EmpName, e.City, d.DeptName, d.Location FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID;

+---------+--------+------------------+----------+

| EmpName | City | DeptName | Location |

+---------+--------+------------------+----------+

| Aarav | Delhi | Computer Science | Block A |

| Priya | Mumbai | Pharmacy | Block B |

| Rahul | Delhi | Computer Science | Block A |

| Sneha | Pune | Management | Block C |

| Vikram | Jaipur | Pharmacy | Block B |

| Kabir | Pune | Management | Block C |

+---------+--------+------------------+----------+

6 rows in set (0.00 sec)

mysql> SELECT e.EmpName, d.DeptName FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID WHERE d.DeptName = 'Computer Science';

+---------+------------------+

| EmpName | DeptName |

+---------+------------------+

| Aarav | Computer Science |

| Rahul | Computer Science |

+---------+------------------+

2 rows in set (0.00 sec)

mysql> SELECT e.EmpName, e.Salary, s.Grade, s.MinSalary FROM Employees e INNER JOIN SalaryGrades s ON e.Salary > s.MinSalary;

+---------+----------+-------+-----------+

| EmpName | Salary | Grade | MinSalary |

+---------+----------+-------+-----------+

| Aarav | 65000.00 | G2 | 50000.00 |

| Aarav | 65000.00 | G1 | 0.00 |

| Priya | 72000.00 | G3 | 70000.00 |

| Priya | 72000.00 | G2 | 50000.00 |

| Priya | 72000.00 | G1 | 0.00 |

| Rahul | 48000.00 | G1 | 0.00 |

| Sneha | 91000.00 | G4 | 90000.00 |

| Sneha | 91000.00 | G3 | 70000.00 |

| Sneha | 91000.00 | G2 | 50000.00 |

| Sneha | 91000.00 | G1 | 0.00 |

| Vikram | 55000.00 | G2 | 50000.00 |

| Vikram | 55000.00 | G1 | 0.00 |

| Neha | 80000.00 | G3 | 70000.00 |

| Neha | 80000.00 | G2 | 50000.00 |

| Neha | 80000.00 | G1 | 0.00 |

| Kabir | 43000.00 | G1 | 0.00 |

+---------+----------+-------+-----------+

16 rows in set (0.00 sec)

```
mysql> SELECT e.EmpName, e.Salary, s.Grade FROM Employees e INNER JOIN SalaryGrades s
ON e.Salary >= s.MinSalary AND e.Salary <= s.MaxSalary;
+---------+----------+-------+
| EmpName | Salary   | Grade |
+---------+----------+-------+
| Aarav   | 65000.00 | G2    |
| Priya   | 72000.00 | G3    |
| Rahul   | 48000.00 | G1    |
| Sneha   | 91000.00 | G4    |
| Vikram  | 55000.00 | G2    |
| Neha    | 80000.00 | G3    |
| Kabir   | 43000.00 | G1    |
+---------+----------+-------+
7 rows in set (0.00 sec)

mysql> SELECT e.EmpName, s.Grade, e.Salary, s.MaxSalary FROM Employees e INNER JOIN
SalaryGrades s ON e.Salary < s.MaxSalary;
+---------+-------+----------+-----------+
| EmpName | Grade | Salary   | MaxSalary |
+---------+-------+----------+-----------+
| Aarav   | G4    | 65000.00 | 120000.00 |
| Aarav   | G3    | 65000.00 |  89999.00 |
| Aarav   | G2    | 65000.00 |  69999.00 |
| Priya   | G4    | 72000.00 | 120000.00 |
| Priya   | G3    | 72000.00 |  89999.00 |
| Rahul   | G4    | 48000.00 | 120000.00 |
| Rahul   | G3    | 48000.00 |  89999.00 |
| Rahul   | G2    | 48000.00 |  69999.00 |
| Rahul   | G1    | 48000.00 |  49999.00 |
```

| Sneha | G4 | 91000.00 | 120000.00 |

| Vikram | G4 | 55000.00 | 120000.00 |

| Vikram | G3 | 55000.00 | 89999.00 |

| Vikram | G2 | 55000.00 | 69999.00 |

| Neha | G4 | 80000.00 | 120000.00 |

| Neha | G3 | 80000.00 | 89999.00 |

| Kabir | G4 | 43000.00 | 120000.00 |

| Kabir | G3 | 43000.00 | 89999.00 |

| Kabir | G2 | 43000.00 | 69999.00 |

| Kabir | G1 | 43000.00 | 49999.00 |

+---------+-------+----------+-----------+

19 rows in set (0.02 sec)

mysql> SELECT e.EmpName, e.Salary, s.Grade FROM Employees e INNER JOIN SalaryGrades s ON e.Salary >= s.MinSalary;

+---------+----------+-------+

| EmpName | Salary | Grade |

+---------+----------+-------+

| Aarav | 65000.00 | G2 |

| Aarav | 65000.00 | G1 |

| Priya | 72000.00 | G3 |

| Priya | 72000.00 | G2 |

| Priya | 72000.00 | G1 |

| Rahul | 48000.00 | G1 |

| Sneha | 91000.00 | G4 |

| Sneha | 91000.00 | G3 |

| Sneha | 91000.00 | G2 |

| Sneha | 91000.00 | G1 |

| Vikram | 55000.00 | G2 |

| Vikram | 55000.00 | G1 |

| Neha | 80000.00 | G3 |

| Neha | 80000.00 | G2 |

| Neha | 80000.00 | G1 |

| Kabir | 43000.00 | G1 |

+---------+----------+-------+

16 rows in set (0.00 sec)

mysql> SELECT EmpName, DeptID, DeptName FROM Employees NATURAL JOIN Departments;

+---------+--------+-----------------+

| EmpName | DeptID | DeptName |

+---------+--------+-----------------+

| Aarav | 10 | Computer Science |

| Priya | 20 | Pharmacy |

| Rahul | 10 | Computer Science |

| Sneha | 30 | Management |

| Vikram | 20 | Pharmacy |

| Kabir | 30 | Management |

+---------+--------+-----------------+

6 rows in set (0.00 sec)

mysql> SELECT EmpName, DeptName, Location FROM Employees NATURAL JOIN Departments WHERE Location = 'Block B';

+---------+----------+----------+

| EmpName | DeptName | Location |

+---------+----------+----------+

| Priya | Pharmacy | Block B |

| Vikram | Pharmacy | Block B |

+---------+----------+----------+

2 rows in set (0.00 sec)

mysql> SELECT EmpName, DeptName, Salary FROM Employees NATURAL JOIN Departments WHERE Salary >= 70000;

+---------+------------+----------+

| EmpName | DeptName | Salary |

+---------+------------+----------+

| Priya | Pharmacy | 72000.00 |

| Sneha | Management | 91000.00 |

+---------+------------+----------+

2 rows in set (0.00 sec)

mysql> SELECT e.EmpName, d.DeptName, e.Salary FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID ORDER BY e.Salary DESC;

+---------+-----------------+----------+

| EmpName | DeptName | Salary |

+---------+-----------------+----------+

| Sneha | Management | 91000.00 |

| Priya | Pharmacy | 72000.00 |

| Aarav | Computer Science | 65000.00 |

| Vikram | Pharmacy | 55000.00 |

| Rahul | Computer Science | 48000.00 |

| Kabir | Management | 43000.00 |

+---------+-----------------+----------+

6 rows in set (0.00 sec)

```
mysql> SELECT DISTINCT d.DeptName FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID;
+------------------+
| DeptName         |
+------------------+
| Computer Science |
| Pharmacy         |
| Management       |
+------------------+
3 rows in set (0.03 sec)

mysql> SELECT d.DeptName, COUNT(e.EmpID) AS TotalEmployees FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID GROUP BY d.DeptName;
+------------------+----------------+
| DeptName         | TotalEmployees |
+------------------+----------------+
| Computer Science |              2 |
| Pharmacy         |              2 |
| Management       |              2 |
+------------------+----------------+
3 rows in set (0.02 sec)

mysql> SELECT d.DeptName, AVG(e.Salary) AS AverageSalary FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID GROUP BY d.DeptName;
+------------------+---------------+
| DeptName         | AverageSalary |
+------------------+---------------+
| Computer Science |  56500.000000 |
| Pharmacy         |  63500.000000 |
```

| Management | 67000.000000 |

+-----------------+--------------+

3 rows in set (0.03 sec)

mysql> SELECT d.DeptName, AVG(e.Salary) AS AverageSalary FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID GROUP BY d.DeptName HAVING AVG(e.Salary) > 60000;

+------------+---------------+

| DeptName | AverageSalary |

+------------+---------------+

| Pharmacy | 63500.000000 |

| Management | 67000.000000 |

+------------+---------------+

2 rows in set (0.02 sec)

mysql> SELECT e.EmpName, d.DeptName, e.Salary FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID ORDER BY e.Salary DESC LIMIT 1;

+---------+------------+----------+

| EmpName | DeptName | Salary |

+---------+------------+----------+

| Sneha | Management | 91000.00 |

+---------+------------+----------+

1 row in set (0.00 sec)

mysql> SELECT e.EmpName, d.DeptName FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID WHERE d.DeptName IN ('Computer Science', 'Pharmacy');

+---------+-----------------+

| EmpName | DeptName |

+---------+-----------------+

| Aarav | Computer Science |

| Priya | Pharmacy |

| Rahul | Computer Science |

| Vikram | Pharmacy |

+---------+------------------+

4 rows in set (0.02 sec)

mysql> SELECT e.EmpName, e.City, e.Salary, d.DeptName FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID WHERE e.City IN ('Delhi', 'Pune') AND e.Salary >= 50000;

+---------+-------+----------+------------------+

| EmpName | City | Salary | DeptName |

+---------+-------+----------+------------------+

| Aarav | Delhi | 65000.00 | Computer Science |

| Sneha | Pune | 91000.00 | Management |

+---------+-------+----------+------------------+

2 rows in set (0.00 sec)

mysql> SELECT e.EmpID, e.EmpName, e.DeptID, d.DeptName FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID;

+-------+---------+--------+------------------+

| EmpID | EmpName | DeptID | DeptName |

+-------+---------+--------+------------------+

| 101 | Aarav | 10 | Computer Science |

| 102 | Priya | 20 | Pharmacy |

| 103 | Rahul | 10 | Computer Science |

| 104 | Sneha | 30 | Management |

| 105 | Vikram | 20 | Pharmacy |

| 107 | Kabir | 30 | Management |

```
+-------+---------+--------+------------------+
```

6 rows in set (0.00 sec)

```
mysql> SELECT DeptName, COUNT(EmpID) AS TotalEmployees FROM Employees NATURAL JOIN Departments GROUP BY DeptName;
+-----------------+----------------+
| DeptName        | TotalEmployees |
+-----------------+----------------+
| Computer Science |              2 |
| Pharmacy        |              2 |
| Management      |              2 |
+-----------------+----------------+
```

3 rows in set (0.00 sec)

```
mysql> SELECT d.DeptName, SUM(e.Salary) AS TotalSalary FROM Employees e INNER JOIN Departments d ON e.DeptID = d.DeptID GROUP BY d.DeptName HAVING SUM(e.Salary) > 100000;
+-----------------+-------------+
| DeptName        | TotalSalary |
+-----------------+-------------+
| Computer Science |   113000.00 |
| Pharmacy        |   127000.00 |
| Management      |   134000.00 |
+-----------------+-------------+
```

3 rows in set (0.02 sec)